

Akhil S Nair

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EDUCATION

University of Illinois Chicago (UIC)

GPA: 4.0/4.0

MS in Computer Science

Aug. 2024 – May 2026

Courses: Computer Algorithms, Machine Learning, Cloud Computing, Android Development, Advanced SWE Topics

APJ Abdul Kalam Technological University

GPA: 7.57/10.0

BTech in Computer Science

Aug. 2017 – June 2021

Courses: Data Structures, Operating Systems, DBMS, Software Engineering, Compiler Design, Computer Networks

SKILLS

Programming Languages: Java, Python, JavaScript, Groovy, C/C++, C#, SQL, Scala, HTML/CSS

Frameworks & Libraries: React.js, Angular, Node.js, Express, Spring Boot, Grails, Material-UI, Tailwind CSS, Passport.js, Jest, Axios, Apollo Client, Apache Spark, Hive, Hadoop, MapReduce, Kafka, TensorFlow

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Git, GitHub, GitLab, Artifactory, CI/CD

Databases: MySQL, Oracle, MongoDB, Postgres, Redis, Elasticsearch

APIs & Protocols: REST, GraphQL, OpenAI Whisper API, Xenova

Developer Tools: VS Code, PyCharm, IntelliJ, Linux, SQS, SDLC, Data Structures & Algorithms

EXPERIENCE

Amazon Web Services (AWS)

May 2025 – Aug 2025

Software Engineering Intern

Seattle, WA

- Contributed to the **Aurora Serverless v2 – Placement Service** team, designing and implementing migration from **DynamoDB**-based configuration management to **CreaMPuff (CMP)**, an **AWS AppConfig**-based RDS solution, reducing manual per-AZ configuration updates by **>90%**.
- Led the integration of **CreaMPuffExtraTastyJavaClient** into the **Java** codebase, creating a new **Guice** module, enabling AZ-aware configuration hierarchy, and executing a safe two-phase (shadow-read → CMP-only) migration, achieving **100% configuration accuracy** in testing.
- Built mismatch detection, deduplication, and cooldown logic for config reads; emitted **CloudWatch metrics** for accuracy tracking and resolved race conditions, cutting duplicate log volume by **≈80%**.
- Delivered a full CMP migration, removing **DynamoDB** dependencies, improving deployment consistency across regions, and enhancing operational efficiency through **source-controlled, pipeline-driven** updates.
- Tech Stack:** Java, AWS AppConfig, TypeScript, AWS CDK, Guice, CloudWatch, DynamoDB.

RxLogix Corporation

Nov 2021 – May 2024

Associate Software Engineer II

India

- Migrated PVCMS backend from GSP to **RESTful services**, improving maintainability and reducing load time by **20%**.
- Resolved **30+ client-reported issues** on the Continuous Engineering team, ensuring system stability and uptime.
- Managed **WHO and KOREA MFDS Dictionary** integration, optimizing data consistency across **Redis**, in-memory storage, and **Elasticsearch**, improving search efficiency by **15%**.
- Utilized **Spring AOP** to design and implement an email intake feature, enabling automated case creation through email parsing and integrating sender acknowledgment emails into the system.

PROJECTS

Hybrid Distributed LLM Architecture ↗ | *Scala, Hadoop, Apache Spark, gRPC, AWS EMR, Amazon Bedrock, Docker*

- Processed 100M+ tokens** by designing and implementing a **cloud-based LLM** trained on the **WikiText-2 dataset**, leveraging **Hadoop MapReduce** for token embeddings and **BPE-based** semantic similarity calculations.
- Cut training time by 35%** using **Apache Spark** and **ND4J** on **AWS EMR**, enabling scalable distributed NLP workloads and applying mathematics/statistics for hyperparameter tuning.
- Reduced deployment time by 50%** via containerized pipelines with **Docker**, and developed a **hybrid conversational system** integrating **AWS Bedrock** and **Ollama** with **gRPC**, **Akka HTTP**, and **API Gateway** for secure, scalable microservices communication.

Assignment Auditor ↗ | *React.js, Material UI, Tailwind CSS, Java 17, Spring Boot, MySQL, Docker, AWS*

- Streamlined assignment reviews by 40%** by building a GitHub-integrated platform with **JWT authentication** via **Spring Security** and API data fetching using **Axios**.
- Achieved 99.9% uptime** by deploying on **AWS EC2** with **Docker**, ensuring scalability and reliability, and **improved reviewer efficiency** with an intuitive **React.js + Material UI** interface for quick status updates.